



30 May 2025

Re. Manuscript submission to Proceedings of the National Academy of Science

Dear Dr Berenbaum,

We are pleased to submit our manuscript entitled "AI-Driven Knowledge Synthesis for Food Web Construction" for consideration for publication in PNAS.

Our manuscript presents a novel methodological framework that leverages Large Language Models (LLMs) to automate the construction of food webs ecosystem models. This addresses a significant bottleneck in ecological modelling, as the traditional approach to constructing these matrices is time-consuming and resource-intensive, often requiring months of expert time. Our framework reduces this process to hours while maintaining ecological validity.

We believe this manuscript is particularly suitable for PNAS due to its broad readership and the extensibility of our approach to a range of domains. Our framework represents a significant methodological innovation in how ecological models can be constructed, introducing a systematic approach to automating a previously manual process through the application of generative AI. The method we describe has immediate practical utility for ecologists and ecosystem modelers, potentially accelerating research in ecosystem-based management and other fields that rely on food web models. We provide rigorous validation of our approach across four distinct marine ecosystems, demonstrating both reproducibility (>99.7% consistency in species grouping decisions) and ecological accuracy (92.6% of taxonomic assignments being at least partially correct). Additionally, the work bridges computational methods and ecological modelling.

Our key findings demonstrate that AI-driven approaches can generate ecologically meaningful components for ecosystem models while dramatically reducing development time. The framework shows strong performance in identifying trophic interactions and estimating diet proportions, with clear pathways for further refinement.

We confirm that this manuscript is not currently under consideration elsewhere, and all authors have approved the submission. We have no conflicts of interest to declare.

Thank you for considering our manuscript. We look forward to your response.

Sincerely,

Scott Spillias and co-authors

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